# Exercise Lecture 10

$$\begin{tikzcd}[row sep=small] & S_u \\ S \arrow[ur, "p"] \arrow[dr, "1-p"'] & \\ & S_d \end{tikzcd}$$

$S = 100$ $\quad p = 1/2$

$S_u = 150$ $\quad r = 0$

$S_d = 75$ $\quad K = 110$

European call

## Real world

$$C_u = \max(S_u - K, 0) = \max(150 - 110, 0) = 40$$

$$C_d = \max(S_d - K, 0) = 0$$

$$C^{Real} = \frac{1}{1+r}\left(p\, C_u + (1-p)\, C_d\right) = 1\left(\frac{1}{2}\cdot 40 + \frac{1}{2}\cdot 0\right) = 20$$

## Risk-neutral

$$q = \frac{S(1+r) - S_d}{S_u - S_d} = \frac{100 - 75}{150 - 75} = \frac{25}{75} = \frac{1}{3}$$

$$C^{Risk\text{-}Neutral} = \left(\frac{1}{3}\cdot 40 + \frac{2}{3}\cdot 0\right) = \frac{40}{3} \approx 13{,}33$$

$$\Rightarrow C^{Real} \neq C^{Risk\text{-}Neutral}$$

Show how you could construct arbitrage strategy if someone would sell you the call at the price computed with real world prob.

$$C^{Real} = 20 > C^{Risk\ neutral}$$

$\Rightarrow$ call is overpriced $\Rightarrow$ sell it and hedge

$$\begin{cases} \alpha^* = \frac{1}{1+r}\left(C_u - \frac{C_u - C_d}{S_u - S_d} S_u\right) \\ \Delta^* = \frac{C_u - C_d}{S_u - S_d} \end{cases}$$

$$\Rightarrow \begin{cases} \Delta^* = \frac{40-0}{150-75} = \frac{40}{75} = \frac{8}{15} \\ \alpha^* = 1\left(40 - \frac{8}{15} \cdot 150\right) = -40 \end{cases}$$

$$\Rightarrow \Pi^* = \alpha^* + \Delta^* S = -40 + \frac{8}{15} \cdot 100$$

$$= 13,33$$

$$= C^{Risk\ neutral}$$

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| | |
|---|---|
| $t = 0$: | CF's |
| Sell at $C^{Real}$ | $+20$ |
| buy $\Delta^*$ shares at 100 | $-53,33$ |
| Borrow $\alpha^* = 40$ | $+40$ |
| | $+6,77$ |

| $t = 1$ | up ($S = 150$) | down ($S = 75$) |
|---|---|---|
| call | $110 - 150 = -40$ | $0$ ($75 < 110$) |
| shares | $\frac{8}{15} \cdot 150 = 80$ | $\frac{8}{15} \cdot 75 = 40$ |
| Repay loan | $-40$ | $-40$ |
| + | $0$ | $0$ |

=> The replicating portfolio cancels your obligation in every scenario and you locked in 6,67 at $t = 0$